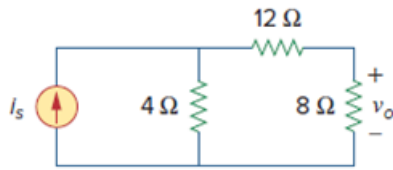


Problem

For the circuit in Fig. 4.3, find v_o when $i_s = 30$ and $i_s = 45$ A.

Figure 4.3:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram shown in Figure 1.

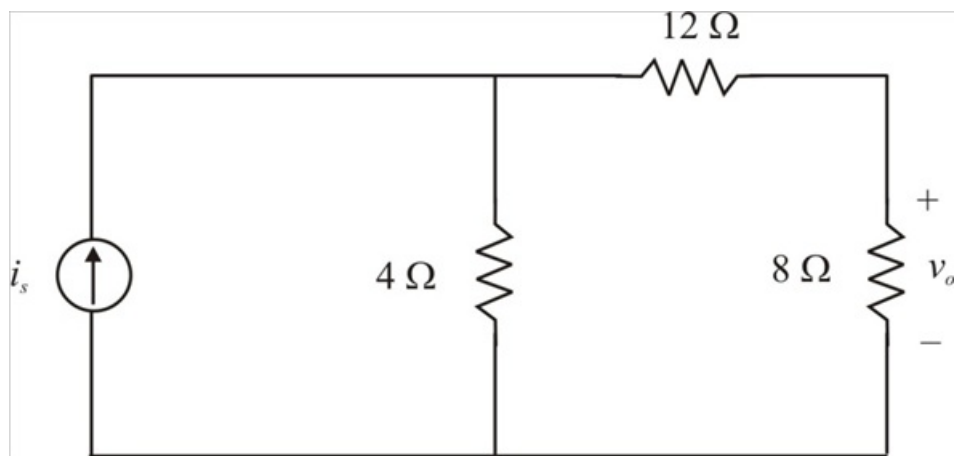


Figure 1

Step 2 of 4

The circuit is labeled as shown in Figure 2.

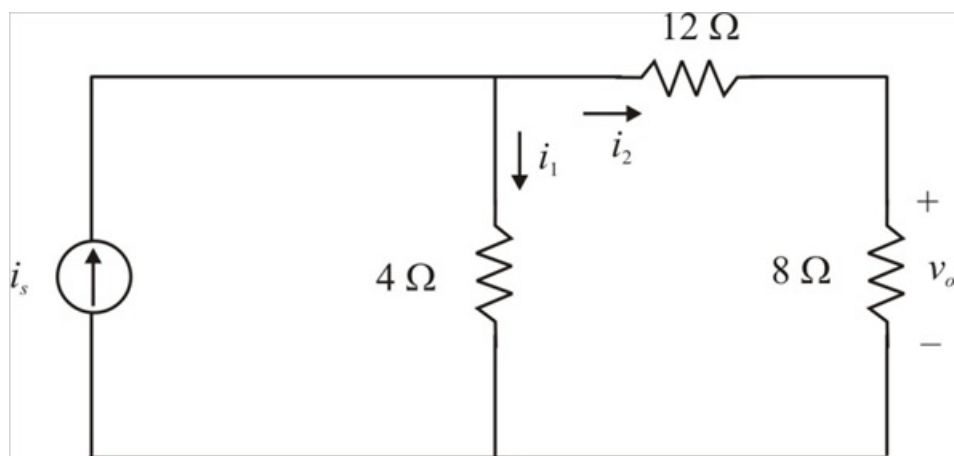


Figure 2

Step 3 of 4

When $i_s = 30\text{ A}$:

By current division, the current i_2 can be calculated as:

$$\begin{aligned} i_2 &= \frac{4}{4+12+8}(i_s) \\ &= \frac{4}{24}(30) \\ &= 5\text{ A} \end{aligned}$$

The voltage v_o can be calculated as:

$$\begin{aligned} v_o &= i_2 R \\ &= (5)(8) \\ &= 40\text{ V} \end{aligned}$$

Therefore the voltage v_o at when $i_s = 30\text{ A}$ is 40 V.

Step 4 of 4

When $i_s = 45\text{ A}$:

By current division, the current i_2 can be calculated as:

$$\begin{aligned} i_2 &= \frac{4}{4+12+8}(i_s) \\ &= \frac{4}{24}(45) \\ &= 7.5\text{ A} \end{aligned}$$

The voltage v_o can be calculated as:

$$\begin{aligned} v_o &= i_2 R \\ &= (7.5)(8) \\ &= 60\text{ V} \end{aligned}$$

Therefore the voltage v_o at when $i_s = 45\text{ A}$ is 60 V.